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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 25th July Evening Shift

JEE Main Question Paper with Answer Key

25 Jul 2021	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	23 questions
Physics	25 questions
Out of syllabus	19 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

In the following the correct bond order sequence is :

(A) $O_2^{2-} > O_2^{+} > O_2^{-} > O_2$

(B) $O_2^{+} > O_2^{-} > O_2^{2-} > O_2$

(C) $O_2^{+} > O_2 > O_2^{-} > O_2^{2-}$

(D) $O_2 > O_2^{-} > O_2^{2-} > O_2^{+}$

Open solution

Q2. MCQ (Single Correct Answer)

Which one of the following metal complexes is most stable?

(A) $[Co(en)(NH_3)_4]Cl_2$

(B) $[Co(en)_3]Cl_2$

(C) $[Co(en)_2(NH_3)_2]Cl_2$

(D) $[Co(NH_3)_6]Cl_2$

Open solution

Q3. MCQ (Single Correct Answer)

The ionic radii of F^- and O^{2-} respectively are 1.33$\overset{o}{A}$ and 1.4$\overset{o}{A}$, while the covalent radius of N is 0.74$\overset{o}{A}$.

The correct statement for the ionic radius of N^{3-} from the following is :

(A) It is smaller than F^- and N

(B) It is bigger than O^{2-} and F^-

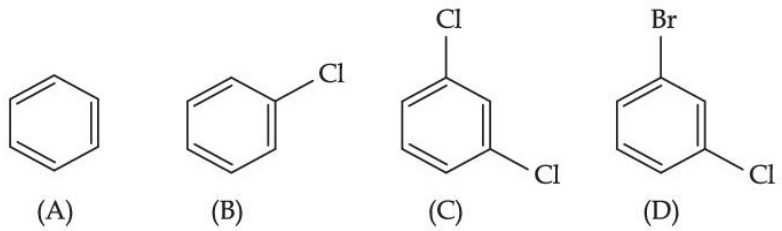
(C) It is bigger than F^- and N, but smaller than of O^{2-}

(D) It is smaller than O^{2-} and F^-, but bigger than of N

Open solution

Q4. MCQ (Single Correct Answer)

The correct decreasing order of densities of the following compounds is :



(A) (B) (C) (D)

(A) (D) > (C) > (B) > (A)

(B) (C) > (D) > (A) > (B)

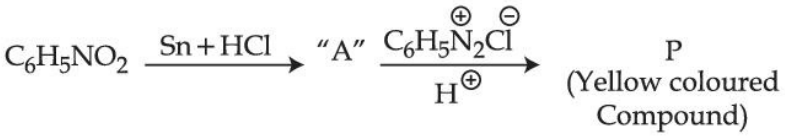
(C) (C) > (B) > (A) > (D)

(D) (A) > (B) > (C) > (D)

Open solution

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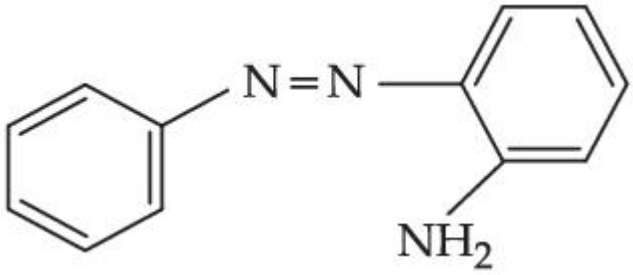
Q5. MCQ (Single Correct Answer)



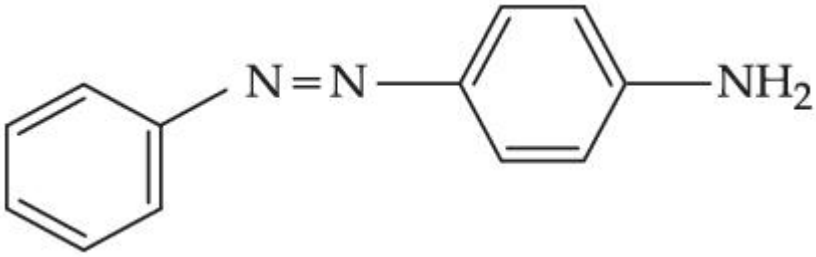
Open solution

Consider the above reaction, the product "P" is :

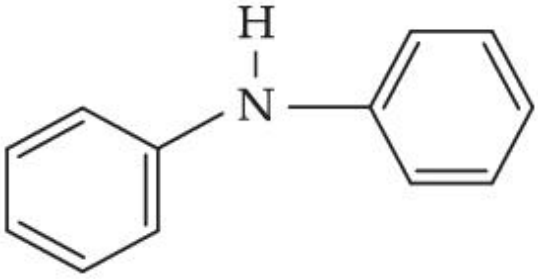
(A)



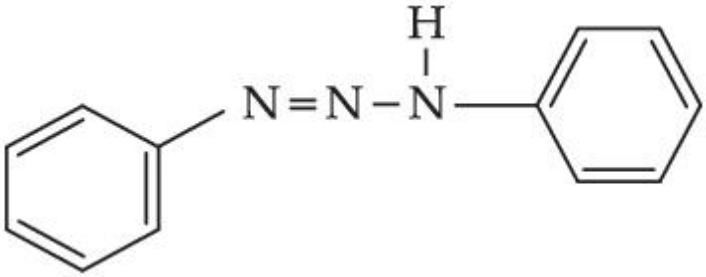
(B)



(C)



(D)



Q6. MCQ (Single Correct Answer)

A reaction of benzonitrile with one equivalent CH_3MgBr followed by hydrolysis produces a yellow liquid "P". The compound "P" will give positive _____.

(A) Iodoform test (B) Schiff's test (C) Ninhydrin's test (D) Tollen's test

Open solution

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Q7. MCQ (Single Correct Answer)

Open solution

The spin only magnetic moments (in BM) for free Ti^{3+}, V^{2+} and Sc^{3+} ions respectively are :

(At.No. Sc : 21, Ti : 22, V : 23)

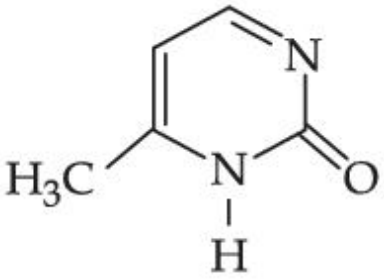
(A) 3.87, 1.73, 0 **(B)** 1.73, 3.87, 0 **(C)** 1.73, 0, 3.87 **(D)** 0, 3.87, 1.73

Q8. MCQ (Single Correct Answer)

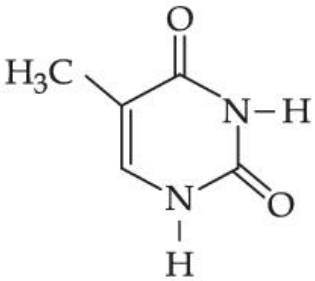
Open solution

Which one of the following is correct structure for cytosine?

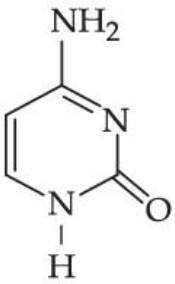
(A)



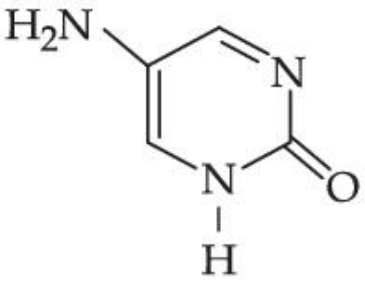
(B)



(C)



(D)



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Q9. MCQ (Single Correct Answer)

Identify the species having one π-bond and maximum number of canonical forms from the following :

(A) SO_3 (B) O_2 (C) SO_2 (D) CO_3^{2-}

Open solution

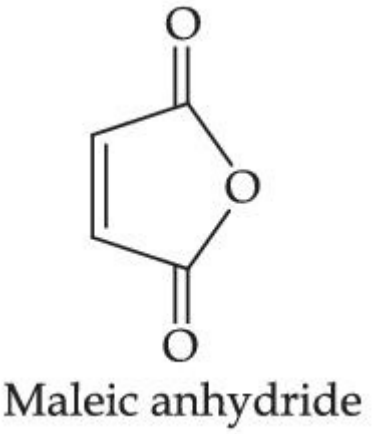
Q10. MCQ (Single Correct Answer)

Which one of the following metals forms interstitial hydride easily?

(A) Cr (B) Fe (C) Mn (D) Co

Open solution

Q11. MCQ (Single Correct Answer)



Maleic anhydride

Maleic anhydride can prepared by :

(A) Heating trans-but-2-enedioic acid

(B) Heating cis-but-2-enedioic acid

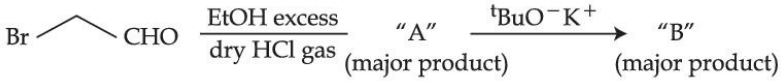
(C) Treating cis-but-2-enedioic acid with alcohol and acid

(D) Treating trans-but-2-enedioic acid with alcohol and acid

Open solution

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Q12. MCQ (Single Correct Answer)

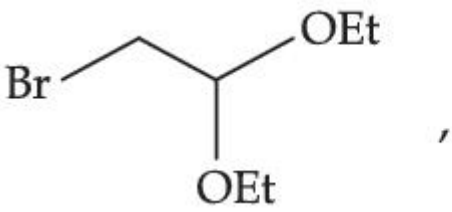


Open solution

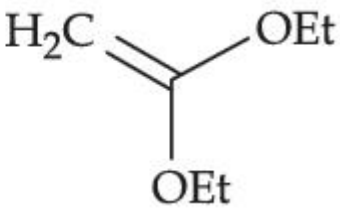
[where $Et \Rightarrow -C_2H_5$, $^tBu \Rightarrow (CH_3)_3C-$]

Consider the above reaction sequence, Product "A" and Product "B" formed respectively are :

(A)



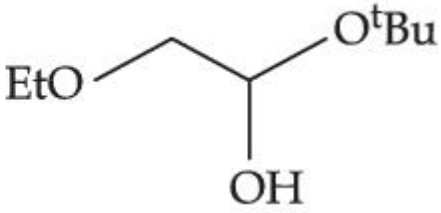
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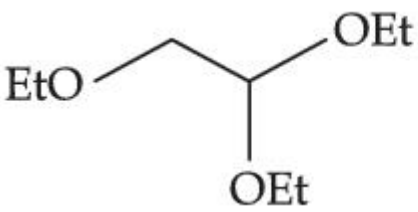
(B)



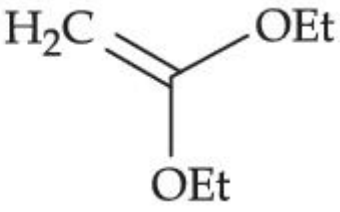
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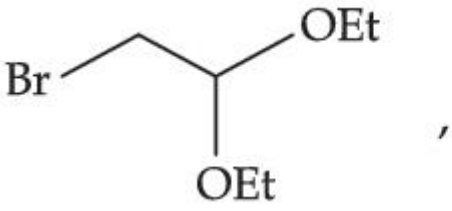
(C)



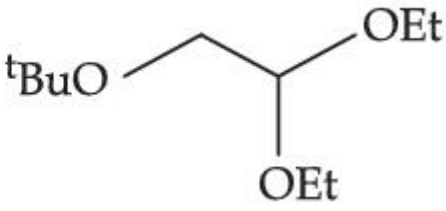
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(D)



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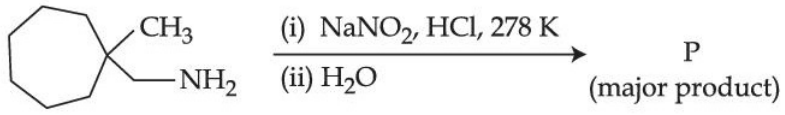


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Q13. MCQ (Single Correct Answer)

Open solution

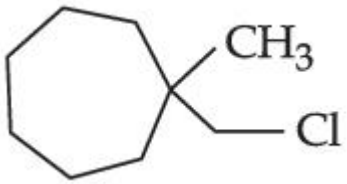
What is the major product "P" of the following reaction?



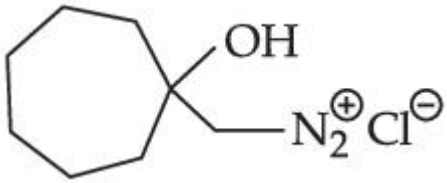
(A)



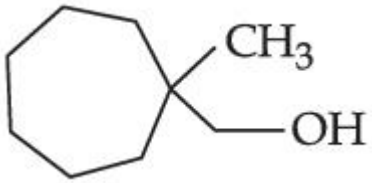
(B)



(C)



(D)



Q14. MCQ (Single Correct Answer)

Open solution

Identify the process in which change in the oxidation state is five :

(A) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

(B) $MnO_4^- \rightarrow Mn^{2+}$

(C) $CrO_4^{2-} \rightarrow Cr^{3+}$

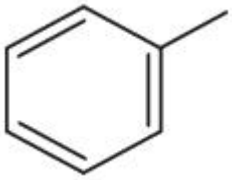
(D) $C_2O_4^{2-} \rightarrow 2CO_2$

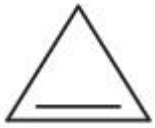
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Q15. MCQ (Single Correct Answer)

Which among the following is the strongest acid ?

(A) $CH_3CH_2CH_2CH_3$

(B) 

(C) 

(D) 

Open solution

Q16. Numerical

A system does 200 J of work and at the same time absorbs 150 J of heat. The magnitude of the change in internal energy is _____ J. (Nearest integer)

Open solution

Q17. Numerical

An accelerated electron has a speed of 5×10^6 ms^{-1} with an uncertainty of 0.02%. The uncertainty in finding its location while in motion is $x \times 10^{-9}$ m. The value of x is _____. (Nearest integer)

[Use mass of electron = 9.1×10^{-31} kg, h =6.63×10^{-34} Js, π = 3.14]

Open solution

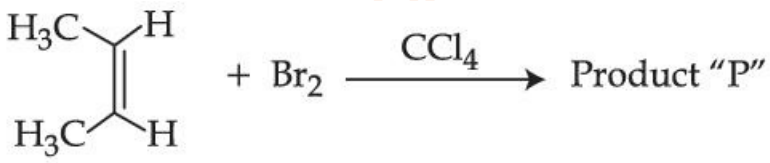
Q18. Numerical

Number of electrons present in 4f orbital of Ho^{3+} ion is _____. (Given Atomic No. of Ho = 67)

Open solution

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Q19. Numerical



Open solution

Consider the above chemical reaction. The total number of stereoisomers possible for Product 'P' is _____.

Q20. Numerical

For a chemical reaction A $\rightarrow$ B, it was found that concentration of B is increased by 0.2 mol L^{-} in 30 min. The average rate of the reaction is _____ $\times 10^{-1}$ mol L^{-1} h^{-1}. (in nearest integer)

Open solution

Q21. Numerical

Assuming that $Ba(OH)_2$ is completely ionised in aqueous solution under the given conditions the concentration of H_3O^+ ions in 0.005 M aqueous solution of $Ba(OH)_2$ at 298 K is _____ $\times 10^{-12}$ mol L^{-1}. (Nearest integer)

Open solution

Q22. Numerical

0.8 g of an organic compound was analyzed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then _____ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis.

Open solution

Q23. Numerical

When 3.00 g of a substance 'X' is dissolved in 100 g of CCl_4, it raises the boiling point by 0.60 K. The molar mass of the substance 'X' is _____ g mol^{-1}. (Nearest integer).

[Given K_b for CCl_4 is 5.0 K kg mol^{-1}]

Open solution

Mathematics 23 questions

Q24. MCQ (Single Correct Answer)

The sum of all those terms which are rational numbers in the

expansion of $(2^{1/3} + 3^{1/4})^{12}$ is :

(A) 89 (B) 27 (C) 35 (D) 43

Open solution

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Q25. MCQ (Single Correct Answer)

The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then the standard deviation of the second sample is :

Open solution

(A) 8 (B) 6 (C) 4 (D) 5

Q26. MCQ (Single Correct Answer)

If $f(x) = \begin{cases} \int_0^x (5+|1-t|)dt, & x > 2 \\ 5x+1, & x \le 2 \end{cases}$, then

Open solution

(A) f(x) is not continuous at x = 2

(B) f(x) is everywhere differentiable

(C) f(x) is continuous but not differentiable at x = 2

(D) f(x) is not differentiable at x = 1

Q27. MCQ (Single Correct Answer)

If the greatest value of the term independent of 'x' in the

expansion of $\left(x\sin\alpha + a\frac{\cos\alpha}{x}\right)^{10}$ is $\frac{10!}{(5!)^2}$, then the value of 'a' is equal to :

Open solution

(A) -1 (B) 1 (C) -2 (D) 2

Q28. MCQ (Single Correct Answer)

The value of $\cot\frac{\pi}{24}$ is :

Open solution

(A) $\sqrt{2}+\sqrt{3}+2-\sqrt{6}$ (B) $\sqrt{2}+\sqrt{3}+2+\sqrt{6}$

(C) $\sqrt{2}-\sqrt{3}-2+\sqrt{6}$ (D) $3\sqrt{2}-\sqrt{3}-\sqrt{6}$

Q29. MCQ (Single Correct Answer)

The lowest integer which is greater

than $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$ is _____.

Open solution

(A) 3 (B) 4 (C) 2 (D) 1

Q30. MCQ (Single Correct Answer)

The value of the

integral $\int_{-1}^{1} \log\left(x+\sqrt{x^2+1}\right)dx$ is :

Open solution

(A) 2 (B) 0 (C) -1 (D) 1

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Q31. MCQ (Single Correct Answer)

Let a, b and c be distinct positive numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ are co-planar, then c is equal to :

(A) $\frac{2}{\frac{1}{a}+\frac{1}{b}}$ (B) $\frac{a+b}{2}$

(C) $\frac{1}{a}+\frac{1}{b}$ (D) $\sqrt{ab}$

Open solution

Q32. MCQ (Single Correct Answer)

If [x] be the greatest integer less than or equal to x,

then $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$ is equal to :

(A) 0 (B) 4 (C) -2 (D) 2

Open solution

Q33. MCQ (Single Correct Answer)

The number of distinct real roots

of $\begin{vmatrix}\sin x & \cos x & \cos x\\ \cos x & \sin x & \cos x\\ \cos x & \cos x & \sin x\end{vmatrix}=0$ in the interval $-\frac{\pi}{4}\le x\le\frac{\pi}{4}$ is :

(A) 4 (B) 1 (C) 2 (D) 3

Open solution

Q34. MCQ (Single Correct Answer)

If $\left|\vec{a}\right|=2, \left|\vec{b}\right|=5$ and $\left|\vec{a}\times\vec{b}\right|$ = 8, then $\left|\vec{a}.\vec{b}\right|$ is equal to :

(A) 6 (B) 4 (C) 3 (D) 5

Open solution

Q35. MCQ (Single Correct Answer)

The number of real solutions of the equation, $x^2-|x|-12=0$ is :

(A) 2 (B) 3 (C) 1 (D) 4

Open solution

Q36. MCQ (Single Correct Answer)

Consider function f : A $\to$ B and g : B $\to$ C (A, B, C $\subseteq$ R) such that $(gof)^{-1}$ exists, then :

(A) f and g both are one-one (B) f and g both are onto

(C) f is one-one and g is onto (D) f is onto and g is one-one

Open solution

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Q37. MCQ (Single Correct Answer)

Open solution

If $P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$, then P^{50} is :

(A) $\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 25 \\ 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$

Q38. MCQ (Single Correct Answer)

Open solution

Let X be a random variable such that the probability function of a distribution is given by $P(X = 0) = \frac{1}{2}, P(X = j) = \frac{1}{3^j}(j = 1, 2, 3, \ldots, \infty)$. Then the mean of the distribution and P(X is positive and even) respectively are :

(A) $\frac{3}{8}$ and $\frac{1}{8}$

(B) $\frac{3}{4}$ and $\frac{1}{8}$

(C) $\frac{3}{4}$ and $\frac{1}{9}$

(D) $\frac{3}{4}$ and $\frac{1}{16}$

Q39. MCQ (Single Correct Answer)

Open solution

If ${}^nP_r = {}^nP_{r+1}$ and ${}^nC_r = {}^nC_{r-1}$, then the value of r is equal to :

(A) 1

(B) 4

(C) 2

(D) 3

Q40. MCQ (Single Correct Answer)

Open solution

Let y = y(x) be the solution of the differential

equation xdy = (y + x^3 cosx)dx with y(π) = 0, then $y\left(\frac{\pi}{2}\right)$ is equal to :

(A) $\frac{\pi^2}{4} + \frac{\pi}{2}$

(B) $\frac{\pi^2}{2} + \frac{\pi}{4}$

(C) $\frac{\pi^2}{2} - \frac{\pi}{4}$

(D) $\frac{\pi^4}{4} - \frac{\pi}{2}$

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Q41. Numerical

Consider the function

Open solution

where P(x) is a polynomial such that P" (x) is always a constant and P(3) = 9. If f(x) is continuous at x = 2, then P(5) is equal to _____.

$$f(x) = \begin{cases} \dfrac{P(x)}{\sin(x-2)}, & x \neq 2 \\ = 7, & x = 2 \end{cases}$$

Q42. Numerical

The equation of a circle is $Re(z^2) + 2(Im(z))^2 + 2Re(z) = 0$, where z = x + iy. A line which passes through the center of the given circle and the vertex of the parabola, $x^2 - 6x - y + 13 = 0$, has y-intercept equal to _____.

Open solution

Q43. Numerical

If $\left(\vec{a} + 3\vec{b}\right)$ is perpendicular to $\left(7\vec{a} - 5\vec{b}\right)$ and $\left(\vec{a} - 4\vec{b}\right)$ is perpendicular to $\left(7\vec{a} - 2\vec{b}\right)$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is _____.

Open solution

Q44. Numerical

If a + b + c = 1, ab + bc + ca = 2 and abc = 3, then the value of $a^4 + b^4 + c^4$ is equal to _____.

Open solution

Q45. Numerical

A fair coin is tossed n-times such that the probability of getting at least one head is at least 0.9. Then the minimum value of n is _____.

Open solution

Q46. Numerical

If the co-efficient of x^7 and x^8 in the expansion of $\left(2 + \frac{x}{3}\right)^n$ are equal, then the value of n is equal to _____.

Open solution

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Physics 25 questions

Q47. MCQ (Single Correct Answer)

The relation between time t and distance x for a moving body is given as $t = mx^2 + nx$, where m and n are constants. The retardation of the motion is : (When v stands for velocity)

(A) $2mv^3$ (B) $2mnv^3$ (C) $2nv^3$ (D) $2n^2v^3$

Open solution

Q48. MCQ (Single Correct Answer)

In a simple harmonic oscillation, what fraction of total mechanical energy is in the form of kinetic energy, when the particle is midway between mean and extreme position.

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Open solution

Q49. MCQ (Single Correct Answer)

A force $\vec{F} = (40\hat{i} + 10\hat{j})N$ acts on a body of mass 5 kg. If the body starts from rest, its position vector $\vec{r}$ at time t = 10 s, will be :

(A) $(100\hat{i} + 400\hat{j})m$ (B) $(100\hat{i} + 100\hat{j})m$ (C) $(400\hat{i} + 100\hat{j})m$ (D) $(400\hat{i} + 400\hat{j})m$

Open solution

Q50. MCQ (Single Correct Answer)

A prism of refractive index μ and angle of prism A is placed in the position of minimum angle of deviation. If minimum angle of deviation is also A, then the value of A in terms of refractive index is

(A) $2\cos^{-1}\left(\frac{\mu}{2}\right)$ (B) $\sin^{-1}\left(\frac{\mu}{2}\right)$

(C) $\sin^{-1}\left(\sqrt{\frac{\mu-1}{2}}\right)$ (D) $\cos^{-1}\left(\frac{\mu}{2}\right)$

Open solution

Q51. MCQ (Single Correct Answer)

Two ions having same mass have charges in the ratio 1 : 2. They are projected normally in a uniform magnetic field with their speeds in the ratio 2 : 3. The ratio of the radii of their circular trajectories is :

(A) 1 : 4 (B) 4 : 3 (C) 3 : 1 (D) 2 : 3

Open solution

Q52. MCQ (Single Correct Answer)

A 10 Ω resistance is connected across $220V - 50$ Hz AC supply. The time taken by the current to change from its maximum value to the rms value is :

(A) 2.5 ms (B) 1.5 ms (C) 3.0 ms (D) 4.5 ms

Open solution

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Q53. MCQ (Single Correct Answer)

A balloon was moving upwards with a uniform velocity of 10 m/s. An object of finite mass is dropped from the balloon when it was at a height of 75 m from the ground level. The height of the balloon from the ground when object strikes the ground was around :

(take the value of g as 10 m/s^2)

Open solution

(A) 300 m (B) 200 m (C) 125 m (D) 250 m

Q54. MCQ (Single Correct Answer)

If q_f is the free charge on the capacitor plates and q_b is the bound charge on the dielectric slab of dielectric constant k placed between the capacitor plates, then bound charge q_b an be expressed as :

Open solution

(A) $q_b = q_f\left(1 - \frac{1}{\sqrt{k}}\right)$ (B) $q_b = q_f\left(1 - \frac{1}{k}\right)$

(C) $q_b = q_f\left(1 + \frac{1}{\sqrt{k}}\right)$ (D) $q_b = q_f\left(1 + \frac{1}{k}\right)$

Q55. MCQ (Single Correct Answer)

Consider a planet in some solar system which has a mass double the mass of earth and density equal to the average density of earth. If the weight of an object on earth is W, the weight of the same object on that planet will be :

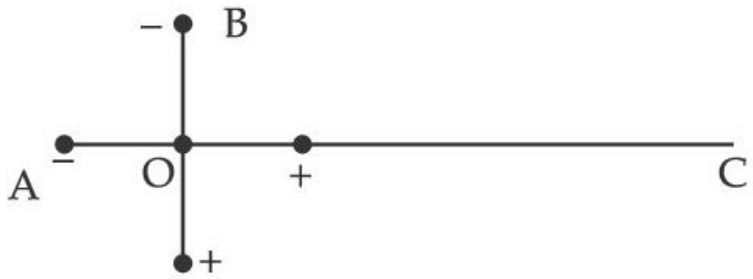
Open solution

(A) 2W (B) W (C) $2\frac{1}{3}$W (D) $\sqrt{2}$W

Q56. MCQ (Single Correct Answer)

Two ideal electric dipoles A and B, having their dipole moment p_1 and p_2 respectively are placed on a plane with their centres at O as shown in the figure. At point C on the axis of dipole A, the resultant electric field is making an angle of 37° with the axis. The ratio of the dipole moment of A and B, $\frac{p_1}{p_2}$ is : (take $\sin 37^\circ = \frac{3}{5}$)

Open solution



(A) $\frac{3}{8}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{4}{3}$

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Q57. MCQ (Single Correct Answer)

Two spherical soap bubbles of radii r_1 and r_2 in vacuum combine under isothermal conditions. The resulting bubble has a radius equal to :

Open solution

(A) $\frac{r_1 r_2}{r_1+r_2}$

(B) $\sqrt{r_1 r_2}$

(C) $\sqrt{r_1^2+r_2^2}$

(D) $\frac{r_1+r_2}{2}$

Q58. MCQ (Single Correct Answer)

The force is given in terms of time t and displacement x by the equation

F = A cos Bx + C sin Dt

Open solution

The dimensional formula of $\frac{AD}{B}$ is :

(A) $[M^0LT^{-1}]$

(B) $[ML^2T^{-3}]$

(C) $[M^1L^1T^{-2}]$

(D) $[M^2L^2T^{-3}]$

Q59. MCQ (Single Correct Answer)

An electron moving with speed v and a photon moving with speed c, have same de-Broglie wavelength. The ratio of kinetic energy of electron to that of photon is :

Open solution

(A) $\frac{3c}{v}$

(B) $\frac{v}{3c}$

(C) $\frac{v}{2c}$

(D) $\frac{2c}{v}$

Q60. MCQ (Single Correct Answer)

The instantaneous velocity of a particle moving in a straight line is given as $V = \alpha t + \beta t^2$, where α and β are constants. The distance travelled by the particle between 1s and 2s is :

Open solution

(A) $3\alpha + 7\beta$

(B) $\frac{3}{2}\alpha + \frac{7}{3}\beta$

(C) $\frac{\alpha}{2} + \frac{\beta}{3}$

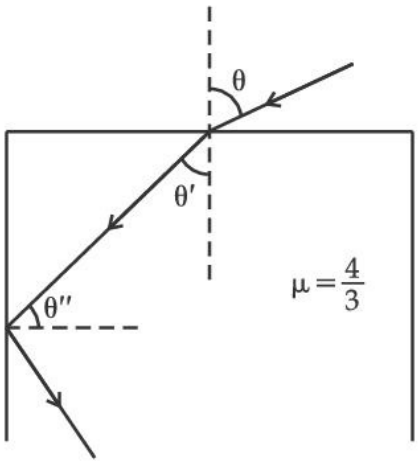
(D) $\frac{3}{2}\alpha + \frac{7}{2}\beta$

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Q61. MCQ (Single Correct Answer)

A ray of light entering from air into a denser medium of refractive index $\frac{4}{3}$, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle θ should be equal to :

Open solution



(A) $\sin^{-1}\frac{\sqrt{7}}{3}$

(B) $\sin^{-1}\frac{\sqrt{5}}{4}$

(C) $\sin^{-1}\frac{\sqrt{7}}{4}$

(D) $\sin^{-1}\frac{\sqrt{5}}{3}$

Q62. MCQ (Single Correct Answer)

When radiation of wavelength λ is incident on a metallic surface, the stopping potential of ejected photoelectrons is 4.8 V. If the same surface is illuminated by radiation of double the previous wavelength, then the stopping potential becomes 1.6 V. The threshold wavelength of the metal is :

Open solution

(A) 2λ (B) 4λ (C) 8λ (D) 6λ

Q63. MCQ (Single Correct Answer)

Two vectors $\vec{X}$ and $\vec{Y}$ have equal magnitude. The magnitude of $(\vec{X}-\vec{Y})$ is n times the magnitude of $(\vec{X}+\vec{Y})$. The angle between $\vec{X}$ and $\vec{Y}$ is :

Open solution

(A) $\cos^{-1}\left(\frac{-n^2-1}{n^2-1}\right)$

(B) $\cos^{-1}\left(\frac{n^2-1}{-n^2-1}\right)$

(C) $\cos^{-1}\left(\frac{n^2+1}{-n^2-1}\right)$

(D) $\cos^{-1}\left(\frac{n^2+1}{n^2-1}\right)$

Q64. Numerical

A system consists of two types of gas molecules A and B having same number density $2 \times 10^{25}/m^3$. The diameter of A and B are 10 $\mathring{A}$ and 5 $\mathring{A}$ respectively. They suffer collision at room temperature. The ratio of average distance covered by the molecule A to that of B between two successive collision is _____ $\times 10^{-2}$

Open solution

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Q65. Numerical

A light beam of wavelength 500 nm is incident on a metal having work function of 1.25 eV, placed in a magnetic field of intensity B. The electrons emitted perpendicular to the magnetic field B, with maximum kinetic energy are bent into circular arc of radius 30 cm. The value of B is _____ $\times 10^{-7}$ T.

Given hc = 20×10^{-26} J-m, mass of electron = 9×10^{-31} kg

Open solution

Q66. Numerical

A 16 Ω wire is bent to form a square loop. A 9V supply having internal resistance of 1Ω is connected across one of its sides. The potential drop across the diagonals of the square loop is _____ $\times 10^{-1}$ V

Open solution

Q67. Numerical

Two circuits are shown in the figure (a) & (b). At a frequency of _____ rad/s the average power dissipated in one cycle will be same in both the circuits.

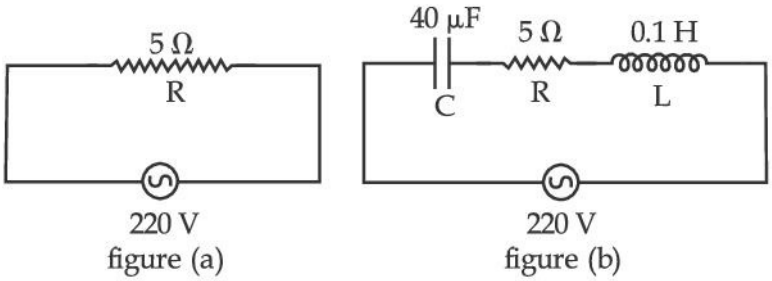


figure (a) figure (b)

Open solution

Q68. Numerical

From the given data, the amount of energy required to break the nucleus of aluminium $^{27}_{13}Al$ is _____ x $\times 10^{-3}$ J.

Mass of neutron = 1.00866 u

Mass of proton = 1.00726 u

Mass of Aluminium nucleus = 27.18846 u

(Assume 1 u corresponds to x J of energy)

(Round off to the nearest integer)

Open solution

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Q69. Numerical

A force of F = (5y + 20)$\hat{j}$ N acts on a particle. The work done by this force when the particle is moved from y = 0 m to y = 10 m is _____ J.

Open solution

Q70. Numerical

A solid disc of radius 20 cm and mass 10 kg is rotating with an angular velocity of 600 rpm, about an axis normal to its circular plane and passing through its centre of mass. The retarding torque required to bring the disc at rest in 10 s is _____ $\pi \times 10^{-1}$ Nm.

Open solution

Q71. Numerical

In a semiconductor, the number density of intrinsic charge carries at 27°C is $1.5 \times 10^{16}/m^3$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22}/m^3$. The electron density in the doped semiconductor is _____ $\times 10^9/m^3$.

Open solution

Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q72. MCQ (Single Correct Answer) · **Chemistry**

A biodegradable polyamide can be made from :

(A) Glycine and isoprene

(B) Hexamethylene diamine and adipic acid

(C) Glycine and aminocaproic acid

(D) Styrene and caproic acid

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Match List I with List II :

	List-I Elements		List-II Properties
(a)	Li	(i)	Poor water solubility of I^- salt
(b)	Na	(ii)	Most abundant element in cell fluid
(c)	K	(iii)	Bicarbonate salt used in fire extinguisher
(d)	Cs	(iv)	Carbonate salt decomposes easily on heating

Open solution

Choose the correct answer from the options given below :

(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(B) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(C) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(D) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II : (Both having metallurgical terms)

	List - I		List - II
(a)	Concentration of Ag ore	(i)	Reverberatory furnace
(b)	Blast furnace	(ii)	Pig iron
(c)	Blister copper	(iii)	Leaching with dilute NaCN solution
(d)	Froth floatation method	(iv)	Sulfide ores

Choose the correct answer from the options given below :

(A) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(D) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : Chlorofluoro breakdown by radiation in the visible energy region and release chlorine gas in the atmosphere which then reacts with stratospheric ozone.

Statement II : Atmospheric ozone reacts with nitric oxide to give nitrogen and oxygen gases, which add to the atmosphere.

For the above statements choose the correct answer from the options given below :

(A) Statement I is incorrect but Statement II is true

(B) Both Statement I and II are false

(C) Statement I is correct but Statement II is false

(D) Both Statement I and II are correct

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II :

List-I Example of colloids	List - II Classification
(a) Cheese	(i) dispersion of liquid in liquid
(b) Pumice stone	(ii) dispersion of liquid in gas
(c) Hair cream	(iii) dispersion of gas in solid
(d) Cloud	(iv) dispersion of liquid in solid

Choose the most appropriate answer from the options given below

(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(B) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(C) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(D) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Q77. Numerical · **Chemistry**

Open solution

The number of significant figures in 0.00340 is _____.

Q78. Numerical · **Chemistry**

Open solution

An LPG cylinder contains gas at a pressure of 300 kPa at 27°. The cylinder can withstand the pressure of 1.2×10^6 Pa. The room in which the cylinder is kept catches fire. The minimum temperature at which the bursting of cylinder will take place is _____° C. (Nearest integer)

Q79. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Consider the statement "The match will be played only if the weather is good and ground is not wet". Select the correct negation from the following :

(A) The match will not be played and weather is not good and ground is wet.

(B) If the match will not be played, then either weather is not good or ground is wet.

(C) The match will be played and weather is not good or ground is wet.

(D) The match will not be played or weather is good and ground is not wet.

Q80. MCQ (Single Correct Answer) · **Mathematics**

Open solution

If a tangent to the ellipse $x^2 + 4y^2 = 4$ meets the tangents at the extremities of it major axis at B and C, then the circle with BC as diameter passes through the point :

(A) $(\sqrt{3}, 0)$ **(B)** $(\sqrt{2}, 0)$ **(C)** $(1, 1)$ **(D)** $(-1, 1)$

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Q81. MCQ (Single Correct Answer) · **Mathematics**

Let the equation of the pair of lines, y = px and y = qx, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is :

(A) $x^2 - 3xy + y^2 = 0$

(B) $x^2 + 4xy - y^2 = 0$

(C) $x^2 + 3xy - y^2 = 0$

(D) $x^2 - 3xy - y^2 = 0$

Open solution

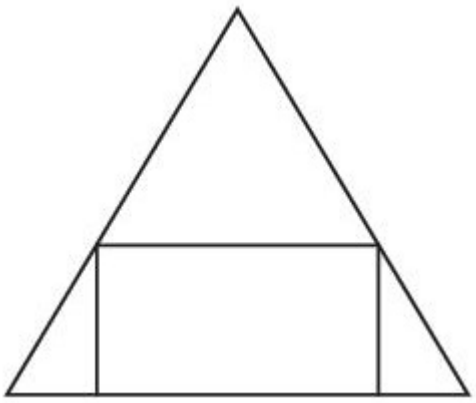
Q82. Numerical · **Mathematics**

Let $n \in N$ and [x] denote the greatest integer less than or equal to x. If the sum of (n + 1) terms ${}^nC_0, 3.{}^nC_1, 5.{}^nC_2, 7.{}^nC_3, \ldots\ldots$ is equal to $2^{100} \cdot 101$, then $2\left[\frac{n-1}{2}\right]$ is equal to _____.

Open solution

Q83. Numerical · **Mathematics**

If a rectangle is inscribed in an equilateral triangle of side length $2\sqrt{2}$ as shown in the figure, then the square of the largest area of such a rectangle is _____.



Open solution

Q84. Numerical · **Mathematics**

Let a curve y = f(x) pass through the point $(2, (\log_e 2)^2)$ and have slope $\frac{2y}{x \log_e x}$ for all positive real value of x. Then the value of f(e) is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

If the lines $\frac{x-k}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x+1}{3} = \frac{y+2}{2} = \frac{z+3}{1}$ are co-planar, then the value of k is _____.

Open solution

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Q86. MCQ (Single Correct Answer) · **Physics**

A heat engine has an efficiency of $\frac{1}{6}$. When the temperature of sink is reduced by 62°C, its efficiency gets doubled. The temperature of the source is :

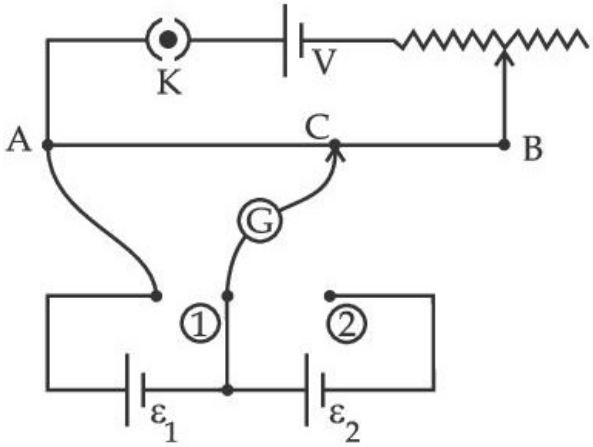
(A) 124°C (B) 37°C (C) 62°C (D) 99°C

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

In the given potentiometer circuit arrangement, the balancing length AC is measured to be 250 cm. When the galvanometer connection is shifted from point (1) to point (2) in the given diagram, the balancing length becomes 400 cm. The ratio of the emf of two cells, $\frac{\varepsilon_1}{\varepsilon_2}$ is :

Open solution

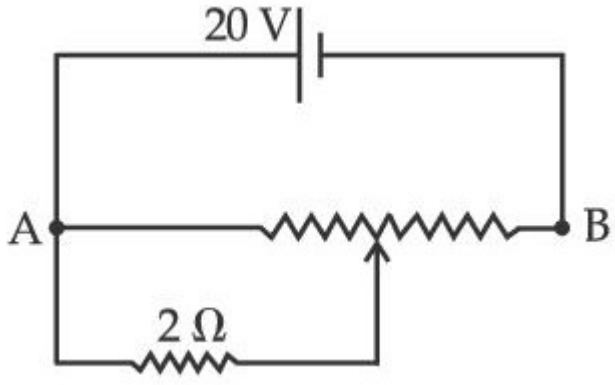


(A) $\frac{5}{3}$ (B) $\frac{8}{5}$ (C) $\frac{4}{3}$ (D) $\frac{3}{2}$

Q88. MCQ (Single Correct Answer) · **Physics**

The given potentiometer has its wire of resistance 10Ω. When the sliding contact is in the middle of the potentiometer wire, the potential drop across 2Ω resistor is :

Open solution



(A) 10 V (B) 5 V (C) $\frac{40}{9}$ V (D) $\frac{40}{11}$ V

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Q89. Numerical · **Physics**

A message signal of frequency 20 kHz and peak voltage of 20 volt is used to modulate a carrier wave of frequency 1 MHz and peak voltage of 20 volt. The modulation index will be :

Open solution

Q90. Numerical · **Physics**

The nuclear activity of a radioactive element becomes $\left(\frac{1}{8}\right)^{th}$ of its initial value in 30 years. The half-life of radioactive element is _____ years.

Open solution

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Answer Key

Chemistry

1. **C**	2. **B**	3. **B**	4. **A**	5. **B**
6. **A**	7. **B**	8. **C**	9. **D**	10. **A**
11. **B**	12. **A**	13. **D**	14. **B**	15. **D**
16. **50**	17. **58**	18. **10**	19. **2**	20. **4**
21. **1**	22. **12**	23. **250**		

Mathematics

24. **D**	25. **C**	26. **C**	27. **D**	28. **B**
29. **A**	30. **B**	31. **D**	32. **B**	33. **B**
34. **A**	35. **A**	36. **C**	37. **A**	38. **B**
39. **C**	40. **A**	41. **39**	42. **1**	43. **60**
44. **13**	45. **4**	46. **55**		

Physics

47. **A**	48. **B**	49. **C**	50. **A**	51. **B**
52. **A**	53. **C**	54. **B**	55. **C**	56. **C**
57. **C**	58. **B**	59. **C**	60. **B**	61. **A**
62. **B**	63. **B**	64. **25**	65. **125**	66. **45**
67. **500**	68. **27**	69. **450**	70. **4**	71. **5**

Out of syllabus

72. **C**	73. **A**	74. **A**	75. **B**	76. **D**
77. **3**	78. **927**	79. **C**	80. **A**	81. **C**
82. **98**	83. **3**	84. **1**	85. **1**	86. **D**
87. **A**	88. **C**	89. **1**	90. **10**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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